How to read a stack trace

```python
from datetime import datetime


def compute_age(birthday):
    # Expects birthday in the form of a tuple
    # (day, month, year)
    year = datetime.now().year
    age = year - birthday[2]
    return age


def print_age(name, birthday):
    print(f"{name} is {compute_age(birthday)} years old")


def main():
    print_age("Andreas", "20-03-1986")


if __name__ == "__main__":
    main()
```



How to read a stack trace

```
Traceback (most recent call last):
  File "/home/andreas/repos/code_examples/stack_trace.py", line 20, in <module>
    main()
  File "/home/andreas/repos/code_examples/stack_trace.py", line 16, in main
    print_age("Andreas", "20-03-1986")
  File "/home/andreas/repos/code_examples/stack_trace.py", line 12, in print_age
    print(f"{name} is {compute_age(birthday)} years old")
                       ^^^^^^^^^^^^^^^^^^^^^
  File "/home/andreas/repos/code_examples/stack_trace.py", line 7, in compute_age
    age = year - birthday[2]
          ~~~~~^~~~~~~~~~~~
TypeError: unsupported operand type(s) for -: 'int' and 'str'
```

```python
from datetime import datetime


def compute_age(birthday):
    # Expects birthday in the for
    # (day, month, year)
    year = datetime.now().year
    age = year - birthday[2]
    return age


def print_age(name, birthday):
    print(f"{name} is {compute_age(birthday)} years old")


def main():
    print_age("Andreas", "20-03-1986")


if __name__ == "__main__":
    main()
```



AALBORG UNIVERSITY
DENMARK

How to read a stack trace

```
Traceback (most recent call last):
  File "/home/andreas/repos/code_examples/stack_trace.py", line 20, in <module>
    main()
  File "/home/andreas/repos/code_examples/stack_trace.py", line 16, in main
    print_age("Andreas", "20-03-1986")
  File "/home/andreas/repos/code_examples/stack_trace.py", line 12, in print_age
    print(f"{name} is {compute_age(birthday)} years old")
                       ^^^^^^^^^^^^^^^^^^^^^
  File "/home/andreas/repos/code_examples/stack_trace.py", line 7, in compute_age
    age = year - birthday[2]
          ~~~~~^~~~~~~~~~~~~
TypeError: unsupported operand type(s) for -: 'int' and 'str'
```

```python
from datetime import datetime


def compute_age(birthday):
    # Expects birthday in the for
    # (day, month, year)
    year = datetime.now().year
    age = year - birthday[2]
    return age


def print_age(name, birthday):
    print(f"{name} is {compute_age(birthday)} years old")


def main():
    print_age("Andreas", "20-03-1986")


if __name__ == "__main__":
    main()
```

What is the error?

How to read a stack trace

```
from datetime import datetime


def compute_age(birthday):
    # Expects birthday in the for
    # (day, month, year)
    year = datetime.now().year
    age = year - birthday[2]
    return age


def print_age(name, birthday):
    print(f"{name} is {compute_age(birthday)} years old")


def main():
    print_age("Andreas", "20-03-1986")


if __name__ == "__main__":
    main()
```

```
Traceback (most recent call last):
  File "/home/andreas/repos/code_examples/stack_trace.py", line 20, in <module>
    main()
  File "/home/andreas/repos/code_examples/stack_trace.py", line 16, in main
    print_age("Andreas", "20-03-1986")
  File "/home/andreas/repos/code_examples/stack_trace.py", line 12, in print_age
    print(f"{name} is {compute_age(birthday)} years old")
                       ^^^^^^^^^^^^^^^^^^^^^
  File "/home/andreas/repos/code_examples/stack_trace.py", line 7, in compute_age
    age = year - birthday[2]
          ~~~~~^~~~~~~~~~~~
TypeError: unsupported operand type(s) for -: 'int' and 'str'
```

Where did the error occur? File, line, function.

How to read a stack trace

```
Traceback (most recent call last):
  File "/home/andreas/repos/code_examples/stack_trace.py", line 20, in <module>
    main()
  File "/home/andreas/repos/code_examples/stack_trace.py", line 16, in main
    print_age("Andreas", "20-03-1986")
  File "/home/andreas/repos/code_examples/stack_trace.py", line 12, in print_age
    print(f"{name} is {compute_age(birthday)} years old")
                       ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/andreas/repos/code_examples/stack_trace.py", line 7, in compute_age
    age = year - birthday[2]
          ~~~~~^~~~~~~~~~~~~
TypeError: unsupported operand type(s) for -: 'int' and 'str'
```

```python
from datetime import datetime


def compute_age(birthday):
    # Expects birthday in the for
    # (day, month, year)
    year = datetime.now().year
    age = year - birthday[2]
    return age


def print_age(name, birthday):
    print(f"{name} is {compute_age(birthday)} years old")


def main():
    print_age("Andreas", "20-03-1986")


if __name__ == "__main__":
    main()
```

Where was the function called from?

How to read a stack trace

```
Traceback (most recent call last):
  File "/home/andreas/repos/code_examples/stack_trace.py", line 20, in <module>
    main()
  File "/home/andreas/repos/code_examples/stack_trace.py", line 16, in main
    print_age("Andreas", "20-03-1986")
  File "/home/andreas/repos/code_examples/stack_trace.py", line 12, in print_age
    print(f"{name} is {compute_age(birthday)} years old")
                       ^^^^^^^^^^^^^^^^^^^^^
  File "/home/andreas/repos/code_examples/stack_trace.py", line 7, in compute_age
    age = year - birthday[2]
          ~~~~~^~~~~~~~~~~~
TypeError: unsupported operand type(s) for -: 'int' and 'str'
```

```python
from datetime import datetime


def compute_age(birthday):
    # Expects birthday in the for
    # (day, month, year)
    year = datetime.now().year
    age = year - birthday[2]
    return age


def print_age(name, birthday):
    print(f"{name} is {compute_age(birthday)} years old")


def main():
    print_age("Andreas", "20-03-1986")


if __name__ == "__main__":
    main()
```

Where was *that* function then called from?



How to read a stack trace

```
Traceback (most recent call last):
  File "/home/andreas/repos/code_examples/stack_trace.py", line 20, in <module>
    main()
  File "/home/andreas/repos/code_examples/stack_trace.py", line 16, in main
    print_age("Andreas", "20-03-1986")
  File "/home/andreas/repos/code_examples/stack_trace.py", line 12, in print_age
    print(f"{name} is {compute_age(birthday)} years old")
                       ^^^^^^^^^^^^^^^^^^^^^
  File "/home/andreas/repos/code_examples/stack_trace.py", line 7, in compute_age
    age = year - birthday[2]
          ~~~~~^~~~~~~~~~~~
TypeError: unsupported operand type(s) for -: 'int' and 'str'
```

```python
from datetime import datetime


def compute_age(birthday):
    # Expects birthday in the for
    # (day, month, year)
    year = datetime.now().year
    age = year - birthday[2]
    return age


def print_age(name, birthday):
    print(f"{name} is {compute_age(birthday)} years old")


def main():
    print_age("Andreas", "20-03-1986")


if __name__ == "__main__":
    main()
```

Where was *that* function then called from?